MTE 203 – Advanced Calculus
Week 5 Homework – Solutions

Higher Order Partial Derivatives and Applications (S.12.5)

Problem 1: [12.5, Prob. 21]

If $z = x^2 + xy + y^2 \sin\left(\frac{x}{y}\right)$, show that

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 2z = x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x\,\partial y} + y^2\frac{\partial^2 z}{\partial y^2}$$

Solution:

Let us first work with the left-hand side (LHS) of the equation:

$$\begin{aligned} x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} &= x\left[2x + y + y\cos\left(\frac{x}{y}\right)\right] + y\left[x + 2y\sin\left(\frac{x}{y}\right) - x\cos\left(\frac{x}{y}\right)\right] \\ &= 2\left[x^2 + xy + y^2\sin\left(\frac{x}{y}\right)\right] = 2f(x,y) \end{aligned}$$

The right-hand side (RHS) is given by,

$$\begin{aligned} x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x\partial y} + y^2\frac{\partial^2 z}{\partial y^2} &= x^2\left[2 - \sin\left(\frac{x}{y}\right)\right] + 2xy\left[1 + \cos\left(\frac{x}{y}\right) + \frac{x}{y}\sin\left(\frac{x}{y}\right)\right] \\ &\quad + y^2\left[2\sin\left(\frac{x}{y}\right) - \frac{2x}{y}\cos\left(\frac{x}{y}\right) - \frac{x^2}{y^2}\sin\left(\frac{x}{y}\right)\right] \\ &= 2x^2 + 2xy + (-x^2 + 2x^2 + 2y^2 - x^2)\sin\left(\frac{x}{y}\right) + (2xy - 2xy)\cos\left(\frac{x}{y}\right) \\ &= 2\left[x^2 + xy + y^2\sin\left(\frac{x}{y}\right)\right] = 2f(x,y) \end{aligned}$$

Problem 2: [12.5, Prob. 27]

A function is said to be a harmonic function in a region R if it satisfies the Laplace's equation in ℝ and has continuous second partial derivatives in ℝ. The Laplace's equation for a function $f x, y, z)$ of three variables is

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

Find a region (if possible) in which the function,

$$f(x,y,z) = \frac{1}{\sqrt{x^2+y^2+z^2}}$$

is harmonic.

Solution:

Let us first verify that $f(x,y,z)$ is harmonic using the Laplace's equation,

From $\dfrac{\partial f}{\partial x} = \dfrac{-x}{(x^2+y^2+z^2)^{3/2}}$, $\dfrac{\partial^2 f}{\partial x^2} = \dfrac{-1}{(x^2+y^2+z^2)^{3/2}} + \dfrac{3x^2}{(x^2+y^2+z^2)^{5/2}} = \dfrac{2x^2-y^2-z^2}{(x^2+y^2+z^2)^{5/2}}$.
With similar results for second derivatives with respect to y and z,

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = \frac{2x^2-y^2-z^2}{(x^2+y^2+z^2)^{5/2}} + \frac{2y^2-x^2-z^2}{(x^2+y^2+z^2)^{5/2}} + \frac{2z^2-x^2-y^2}{(x^2+y^2+z^2)^{5/2}} = 0.$$

Since second partial derivatives are continuous except at $(0,0,0)$, the function is harmonic in any region not containing $(0,0,0)$.

Problem 3: [12.5, Prob. 49]

The Laplace's equation for a function $u(x,y)$ of two variables is

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

a. Show that the following function, is harmonic in the entire xy-plane:

$$u(x,y) = e^x \cos y + x.$$

b. In complex variable theory, two functions $u(x,y)$ and $v(x,y)$ are said to be *harmonic conjugates* in a region R if in R they satisfy the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

Find a function $v(x,y)$ so that the function $u(x,y)$ and $v(x,y)$ are harmonic conjugates.

<u>**Solution**</u>

a. Since

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^x \cos y - e^x \cos y = 0$$

$u(x,y)$ is a harmonic function in the entire $x-y$ plane.

b. To find the harmonic conjugate $v(x,y)$ we know that

$$\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = e^x \cos y + 1 \qquad (1)$$

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = e^x \sin y \qquad (2)$$

We can then reconstruct v by integrating the equations above

From (1),

$$\int v\,dv = \int_{x=constant} (e^x \cos y + 1)dy$$

$$v(x,y) = e^x \sin y + y + \phi(x)$$

and from (2),

$$\int v\,dv = \int_{y=constant} (e^x \sin y)dy$$

$$v(x,y\,) = e^x \sin y + \eta(y)$$

To determine the functions $\phi(x)$, and $\eta(y)$, we equate both equations so that,

$$e^x \sin y + y + \phi(x) = e^x \sin y + \eta(y)$$

and notice that the above equality holds for

$$\eta(y) = y$$

$$\phi(x) = 0.$$

Thus, the function,

$$v(x,y) = e^x \sin y + y$$

is the harmonic conjugate of $u(x,y)$.

Chain Rule for Partial Derivatives (S.12.6.)

Problem 1: [12.6, Prob. 13]

Find the derivative below:

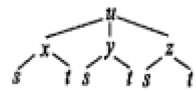
$$\left.\frac{\partial^2 u}{\partial s^2}\right|_t$$

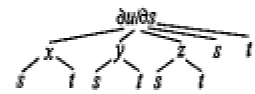
If $u = x^2 + y^2 + z^2 + xyz,\ x = s^2 + t^2,\ y = s^2 - t^2,$ and $z = st$

Solution:

From $\dfrac{\partial u}{\partial s} = \dfrac{\partial u}{\partial x}\dfrac{\partial x}{\partial s} + \dfrac{\partial u}{\partial y}\dfrac{\partial y}{\partial s} + \dfrac{\partial u}{\partial z}\dfrac{\partial z}{\partial s}$

$$= (2x + yz)(2s) + (2y + xz)(2s) + (2z + xy)(t)$$
$$= 2s(2x + 2y + xz + yz) + t(2z + xy),$$





$$\frac{\partial^2 u}{\partial s^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial s}\right)\frac{\partial x}{\partial s} + \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial s}\right)\frac{\partial y}{\partial s} + \frac{\partial}{\partial z}\left(\frac{\partial u}{\partial s}\right)\frac{\partial z}{\partial s} + \frac{\partial}{\partial s}\left(\frac{\partial u}{\partial s}\right)_{x,y,z,t}$$
$$= (4s + 2sz + ty)(2s) + (4s + 2sz + tx)(2s) + (2sx + 2sy + 2t)(t) + 2(2x + 2y + xz + yz)$$
$$= 8s^2(z + 2) + 2(x + y)(2st + z + 2) + 2t^2.$$

Note: Another way to approach this problem is to replace first the values of x, y and z as a function of s and t on $\frac{\partial u}{\partial s}$ before calculating $\frac{\partial^2 u}{\partial s^2}$. The second tree in this case will resemble the first tree, but the function at the top will be $\frac{\partial u}{\partial s}$.

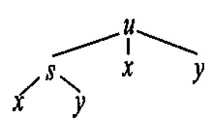
Problem 2: [12.6, Prob. 27]

If $f(s)$ is a differentiable function, show that $u(x, y) = f(4x - 3y) + 5(y - x)$, satisfies the equation

$$3\,\frac{\partial u}{\partial x} + 4\,\frac{\partial u}{\partial y} = 5$$

Solution:

The tree diagram in this case can be drawn as,



If we set $s = 4x - 3y$, then

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial s}\frac{\partial s}{\partial x} + \left.\frac{\partial u}{\partial x}\right)_{s,y} = f'(s)(4) - 5, \quad \frac{\partial u}{\partial y} = \frac{\partial u}{\partial s}\frac{\partial s}{\partial y} + \left.\frac{\partial u}{\partial y}\right)_{s,x} = f'(s)(-3) + 5.$$

Note that the $\frac{\partial u}{\partial x}$ (or $\frac{\partial u}{\partial y}$) on the LHS of both of these equations is not the same as the $\frac{\partial u}{\partial x}$ (or $\frac{\partial u}{\partial y}$) on the RHS.

Hence, $3\dfrac{\partial u}{\partial x} + 4\dfrac{\partial u}{\partial y} = 3[4f'(s) - 5] + 4[-3f'(s) + 5] = 5.$

Problem 3: [12.6, Prob. 35]

An observer travels along the curve $x = t^2,\ y = 3t^3 + 1, z = 2t + 5$, where x, y and z are in meters and $t \geq 0$ is in seconds.

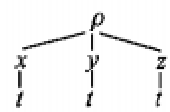
If the density ρ of a gas (in kg/m^3) is given by

$$\rho = \frac{(3x^2 + y^2)}{z^2 + 5}$$

Find the time rate of change of the density of the gas as measured by the observer when $t = 2$ seconds.

Solution:

$$\frac{d\rho}{dt} = \frac{\partial \rho}{\partial x}\frac{dx}{dt} + \frac{\partial \rho}{\partial y}\frac{dy}{dt} + \frac{\partial \rho}{\partial z}\frac{dz}{dt}$$
$$= \left(\frac{6x}{z^2+5}\right)(2t) + \left(\frac{2y}{z^2+5}\right)(9t^2) + \left[-\frac{2z(3x^2+y^2)}{(z^2+5)^2}\right](2).$$



When $t = 2$, we obtain $x = 4$, $y = 25$, $z = 9$, and

$$\frac{d\rho}{dt} = \left(\frac{24}{86}\right)(4) + \left(\frac{50}{86}\right)(36) + \left[-\frac{2(9)(673)}{86^2}\right](2) = 18.77 \text{ kg/m}^3\text{/s}.$$

Directional Derivatives (S.12.8)

Problem 1: [12.8, Prob. 9]

Find the rate of change of $f(x, y) = 2x - 3y$ at (1,1) along the curve $y = x^2$, in the direction of increasing x.

Solution:

The distance traveled along the curve is indicated by the unit tangent vector $\hat{T}$ of the curve at the point $(1,1)$.

To find $\hat{T}$ we need to parametrize the curve. Since we need the direction of increasing x, we can use,

$x = t,$

$y = x^2 \rightarrow y = t^2$

The vector function describing the curve of intersection is then,

$\vec{r}(t) = (t, t^2), t \in R$

The velocity is then given by,

$$\frac{\partial \vec{r}}{\partial t} = (1{,}2t) \left|\frac{\partial \vec{r}}{\partial t}\right| = \sqrt{1 + 4t^2}$$

And the unit tangent vector is,

$$\hat{T} = \frac{(t, 2t)}{\sqrt{1 + 4t^2}}$$

The value of the parameter t at $(1{,}1)$ can be calculated from,

$$x = t = 1$$

$$y = t^2 = 1$$

Therefore $t = 1$ at $(1{,}1)$. We can now evaluate $\hat{T}$ at $t = 1$,

$$\hat{T}\Big|_{t=1} = \frac{(1{,}2)}{\sqrt{1+4}} = \frac{(1{,}2)}{\sqrt{5}}$$

The required rate of change is therefore,

$$D_{\mathbf{T}} f = \nabla f_{|(1,1)} \cdot \hat{\mathbf{T}} = (2, -3) \cdot \frac{(1, 2)}{\sqrt{5}} = \frac{-4}{\sqrt{5}}.$$

Problem 2: [12.8, Prob. 15]

Find the direction in which the function below increases most rapidly at the point $(1, -3{,}2)$. What is the rate of change in that direction?

$$f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$$

Solution:

The function increases more rapidly in the direction of the gradient at the point $(1, -3{,}2)$. The gradient is given by,

$$\vec{\nabla} f = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z})$$

$$\nabla f_{|(1,-3,2)} = \left[\frac{1}{(x^2 + y^2 + z^2)^{3/2}}(-x, -y, -z)\right]_{|(1,-3,2)} = \frac{1}{14\sqrt{14}}(-1, 3, -2),$$

The rate of change in this direction is given by the magnitude of the gradient at that point,

$$\left|\vec{\nabla} f\right| = \frac{1}{14\sqrt{14}}\sqrt{1+9+14} = \frac{1}{14}$$

Problem 3: [12.8, Prob. 25]

Find points on the curve given by

$$C: x = t, y = 1 - 2t, z = t$$

at which the rate of change of the function

$$f(x, y, z) = x^2 + xyz$$

with respect to the distance travelled along the curve vanishes.

Solution:

The distance travelled along a curve is indicated by the unit tangent vector $\hat{T}$. The curve given is a line, hence the unit tangent vector along the line will be given by the unit vector in the direction of the line.

From the parametric equations, the direction of the line is $(1, -2, 1)$. Hence the unit tangent vector is given by,

$$\hat{T} = \frac{(1, -2, 1)}{\sqrt{6}}$$

The rate of change of $f(x, y, z)$ with respect to the distance travelled along a curve vanishes if the directional derivative (rate of change) in the direction of the unit tangent vector is zero.

Hence,

$$0 = D_{\mathbf{T}} f = \nabla f \cdot \hat{\mathbf{T}} = (2x + yz, xz, xy) \cdot \frac{(1, -2, 1)}{\sqrt{6}} \quad \Longrightarrow \quad 0 = 2x + yz - 2xz + xy.$$

If we substitute the parametric equations of the line into this equation,

$$0 = 2t + (1-2t)(t) - 2t(t) + t(1-2t) = -6t^2 + 4t = 2t(2-3t) \quad \Longrightarrow \quad t = 0 \text{ or } t = 2/3.$$

The points on the curve at which the rate of change vanishes can then be calculated by replacing these two values of t in the equation of the line,

$$t = 0 \quad \rightarrow \quad (0,1,0)$$

$$t = \frac{2}{3} \quad \rightarrow \quad (\frac{2}{3}, -\frac{1}{3}, \frac{2}{3})$$

More practice problems[1]

The answers to these problems, along with all even-numbered exercises, are provided at the back of the textbook. Detailed solutions can be found in Trim's Student Solution Manual. If you don't own a copy, a reserved edition is available at the Davis Centre.

Warm-Up Problems

1. S. 12.5, Probs. 12, 16
2. S. 12.6, Probs. 2, 4, 12
3. S. 12.8, Probs. 2, 4, 10, 14

Extra Practice Problems

1. S. 12.5, Probs. 22, 30
2. S. 12.6, Probs. 18, 26
3. S. 12.8, Probs. 18, 20, 30

Extra Challenge Problems

1. S. 12.5, Prob. 36
2. S. 12.6, Prob. 44
3. S. 12.8, Prob. 34

[1] You are strongly encouraged to try and solve the problems assigned in this homework on your own, before looking at the solutions. These problems are only the minimum necessary to master the material. You are also encouraged to do additional problems from the text for practice (some suggested **extra practice problems** have been listed here).